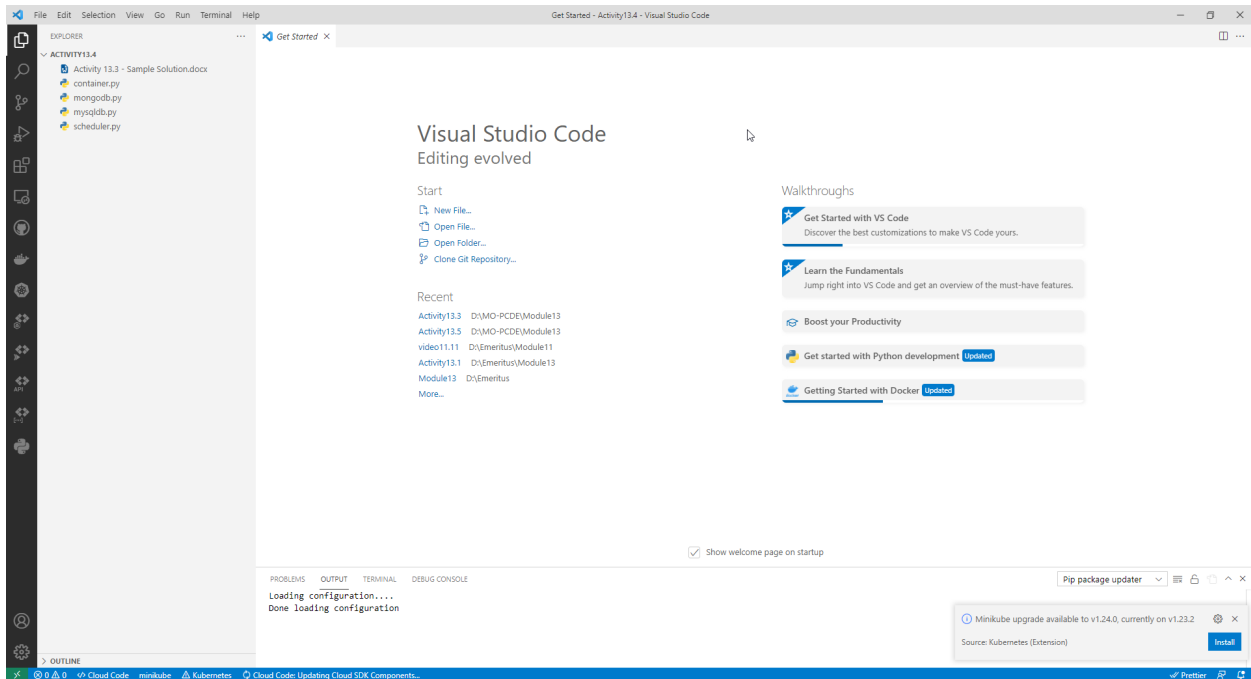
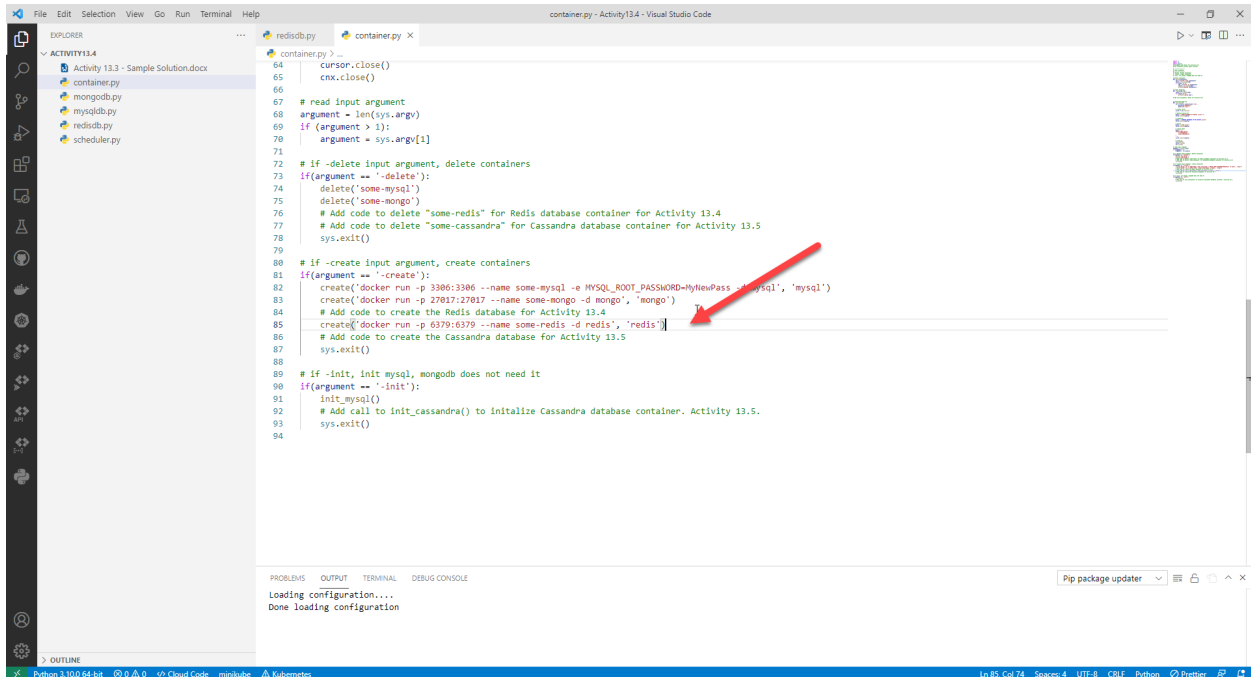


Activity 13.4 Solution File

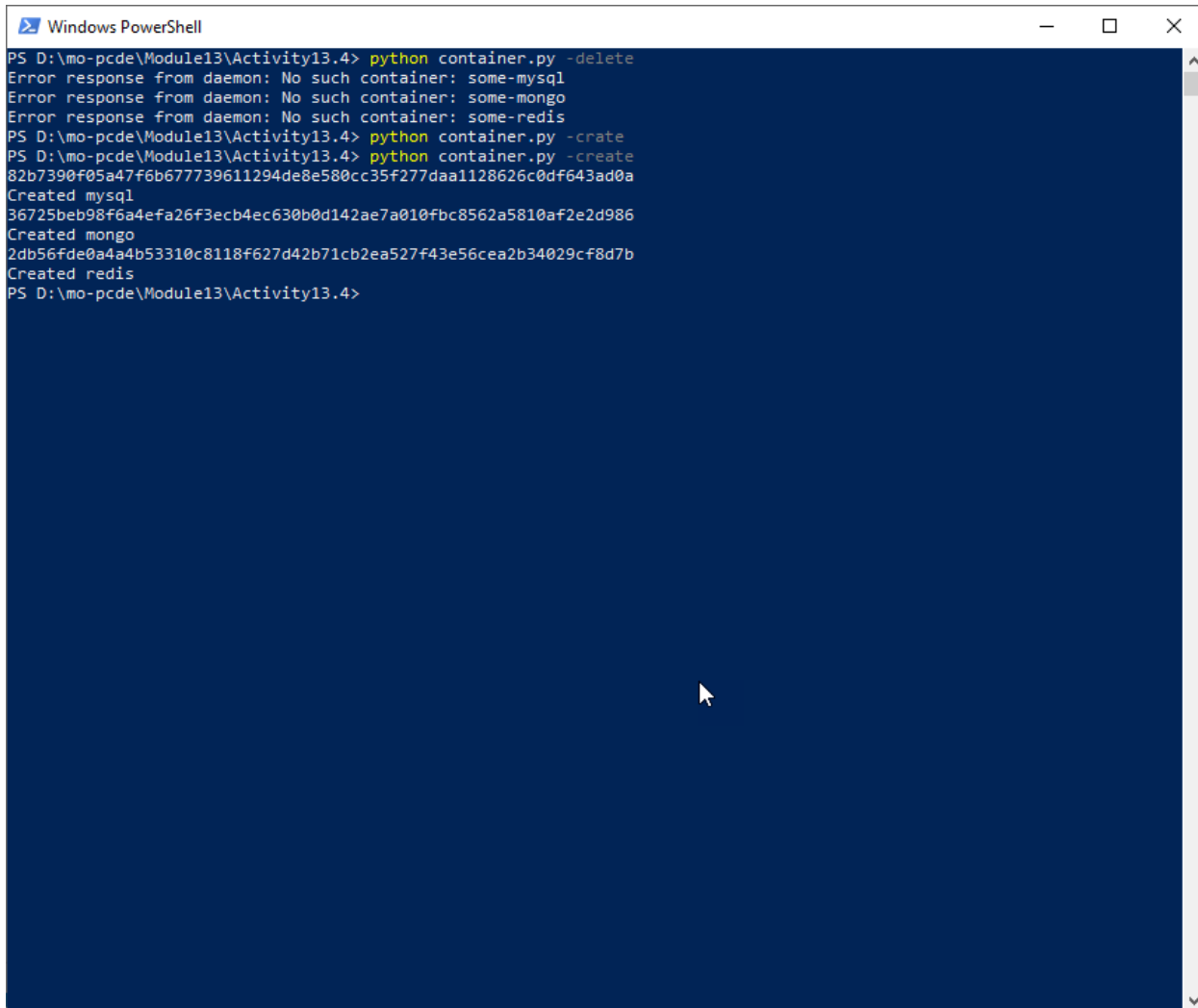
1.



2.

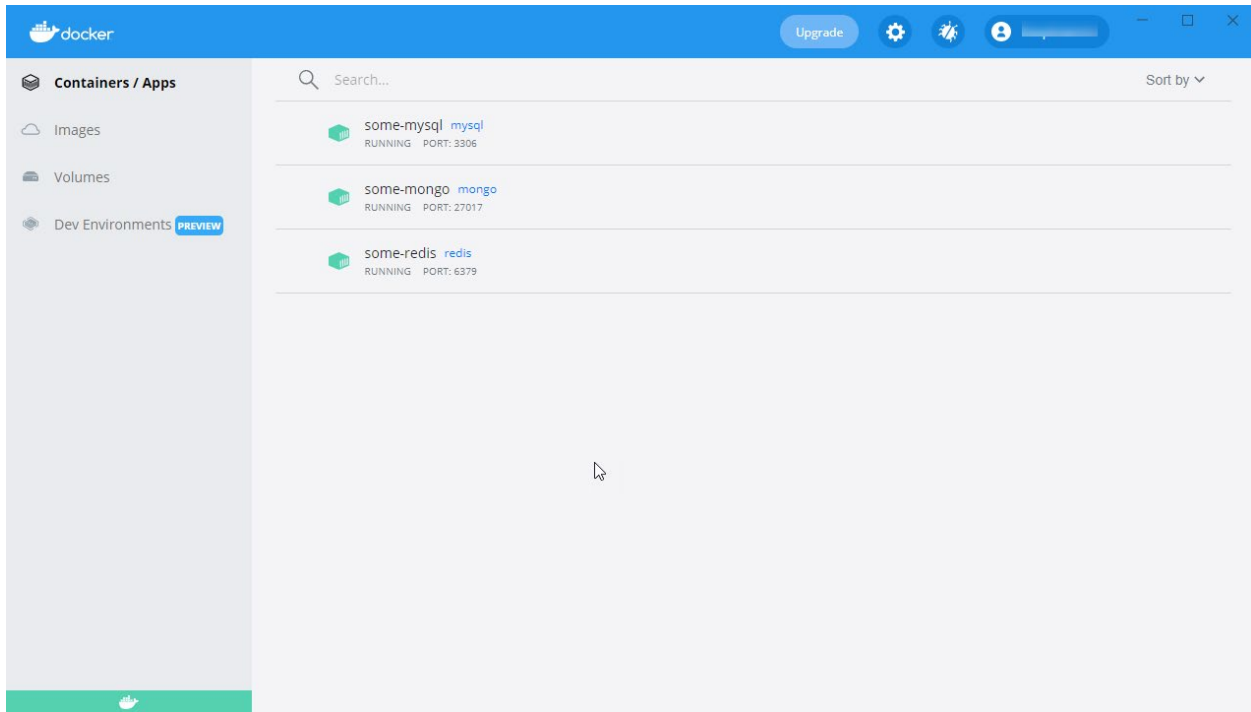


3.

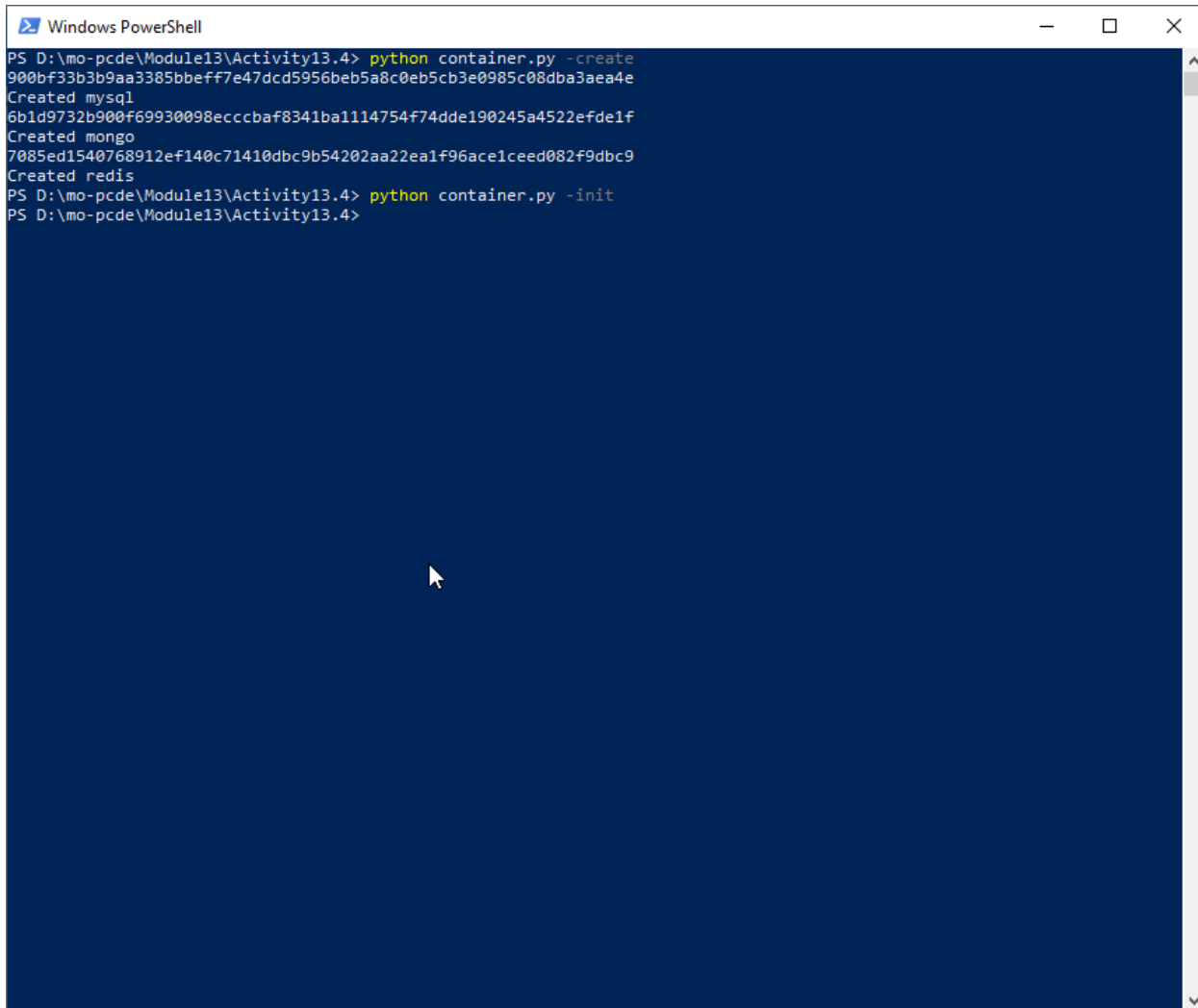


```
Windows PowerShell
PS D:\mo-pcde\Module13\Activity13.4> python container.py -delete
Error response from daemon: No such container: some-mysql
Error response from daemon: No such container: some-mongo
Error response from daemon: No such container: some-redis
PS D:\mo-pcde\Module13\Activity13.4> python container.py -create
PS D:\mo-pcde\Module13\Activity13.4> python container.py -create
82b7390f05a47f6b677739611294de8e580cc35f277daa1128626c0df643ad0a
Created mysql
36725beb98f6a4efa26f3ecb4ec630b0d142ae7a010fbc8562a5810af2e2d986
Created mongo
2db56fde0a4a4b53310c8118f627d42b71cb2ea527f43e56cea2b34029cf8d7b
Created redis
PS D:\mo-pcde\Module13\Activity13.4>
```

4.

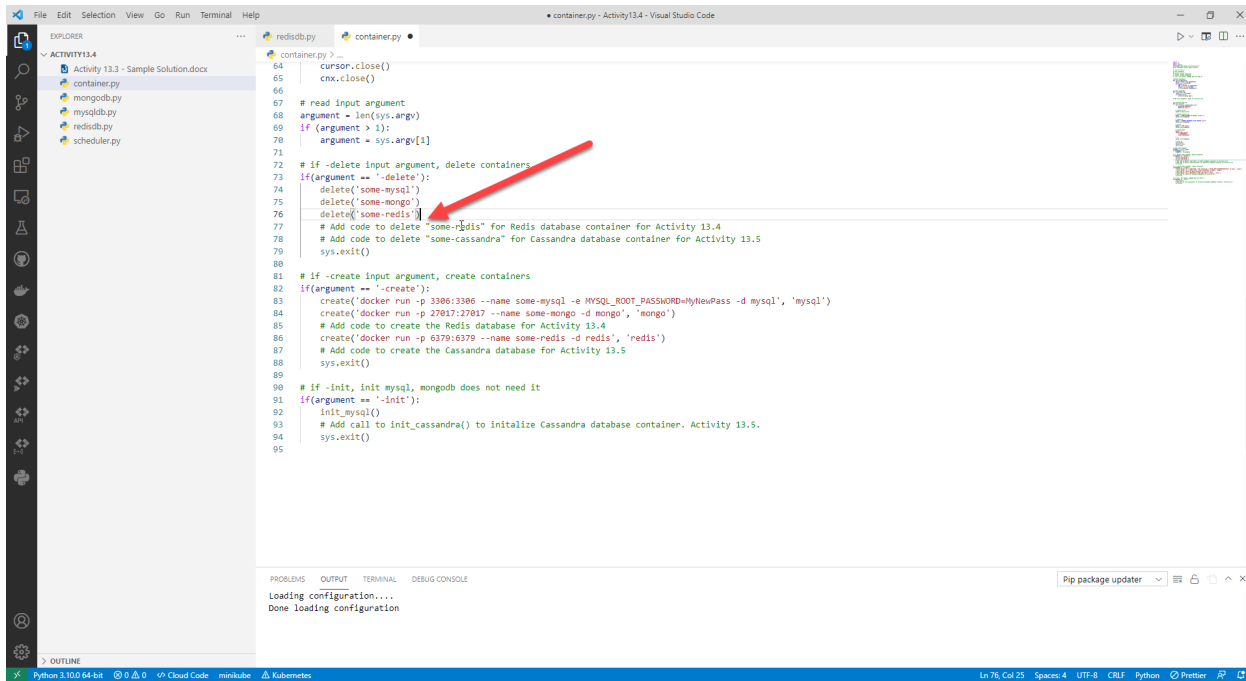


5.



```
Windows PowerShell
PS D:\mo-pcde\Module13\Activity13.4> python container.py -create
900bf33b3b9aa3385bbeff7e47dcd5956beb5a8c0eb5cb3e0985c08dba3aea4e
Created mysql
6b1d9732b900f69930098ecccba8341ba1114754f74dde190245a4522efde1f
Created mongo
7085ed1540768912ef140c71410dbc9b54202aa22ea1f96ace1ceed082f9dbc9
Created redis
PS D:\mo-pcde\Module13\Activity13.4> python container.py -init
PS D:\mo-pcde\Module13\Activity13.4>
```

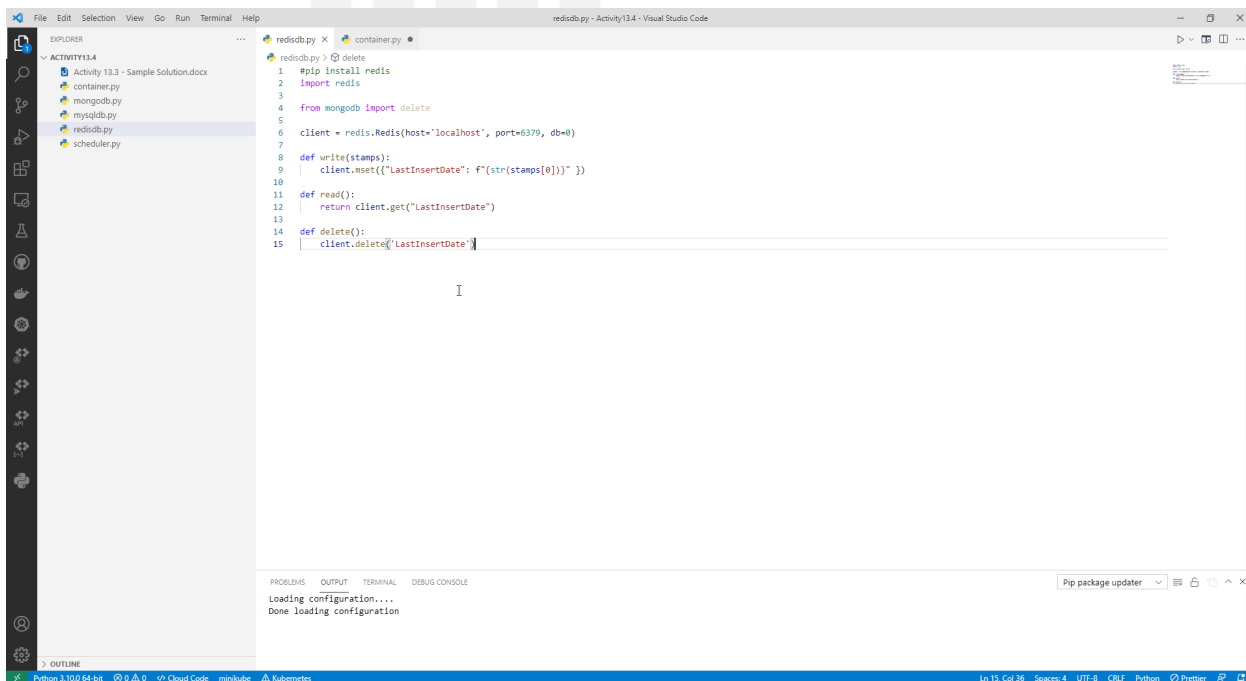
6.



```

64 cursor.close()
65 cnx.close()
66
67 # read input argument
68 argument = len(sys.argv)
69 if (argument > 1):
70     argument = sys.argv[1]
71
72 # if -delete input argument, delete containers
73 if (argument == '-delete'):
74     delete('some-mysql')
75     delete('some-mongo')
76     delete('some-redis')
77
78 # Add code to delete "some-redis" for Redis database container for Activity 13.4
79 # Add code to delete "some-cassandra" for Cassandra database container for Activity 13.5
80 sys.exit()
81
82 # if -create input argument, create containers
83 if (argument == '-create'):
84     create('docker run -p 3306:3306 --name some-mysql -e MYSQL_ROOT_PASSWORD=HyflewPass -d mysql', 'mysql')
85     create('docker run -p 27017:27017 --name some-mongo -d mongo', 'mongo')
86     # Add code to create the Redis database for Activity 13.4
87     create('docker run -p 6379:6379 --name some-redis -d redis', 'redis')
88     # Add code to create the Cassandra database for Activity 13.5
89     sys.exit()
90
91 # if -init, init mysql, mongodb does not need it
92 if (argument == '-init'):
93     init_mysql()
94     # Add call to init_cassandra() to initialize Cassandra database container. Activity 13.5.
95     sys.exit()
  
```

7.

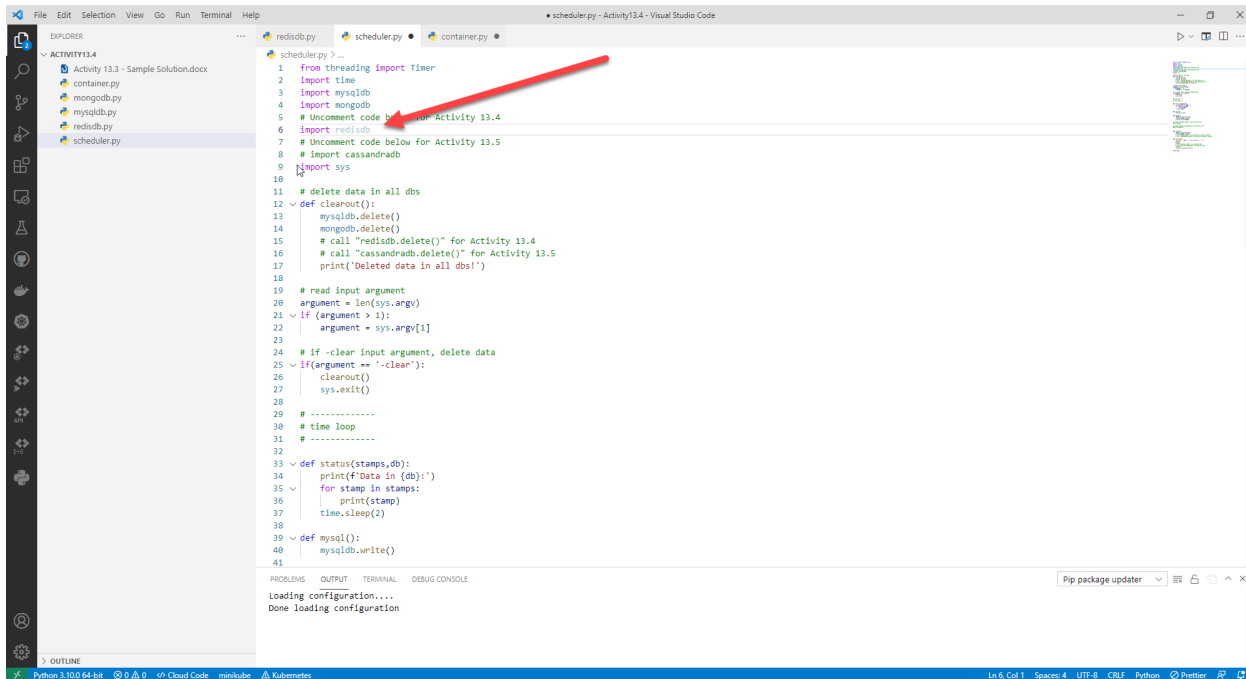


```

1 #pip install redis
2 import redis
3
4 from mongodb import delete
5
6 client = redis.Redis(host='localhost', port=6379, db=0)
7
8 def write(stamps):
9     client.set("LastInsertDate": f'{str(stamps[0])}')
10
11 def read():
12     return client.get("LastInsertDate")
13
14 def delete():
15     client.delete("LastInsertDate")
  
```

Step 8

a.

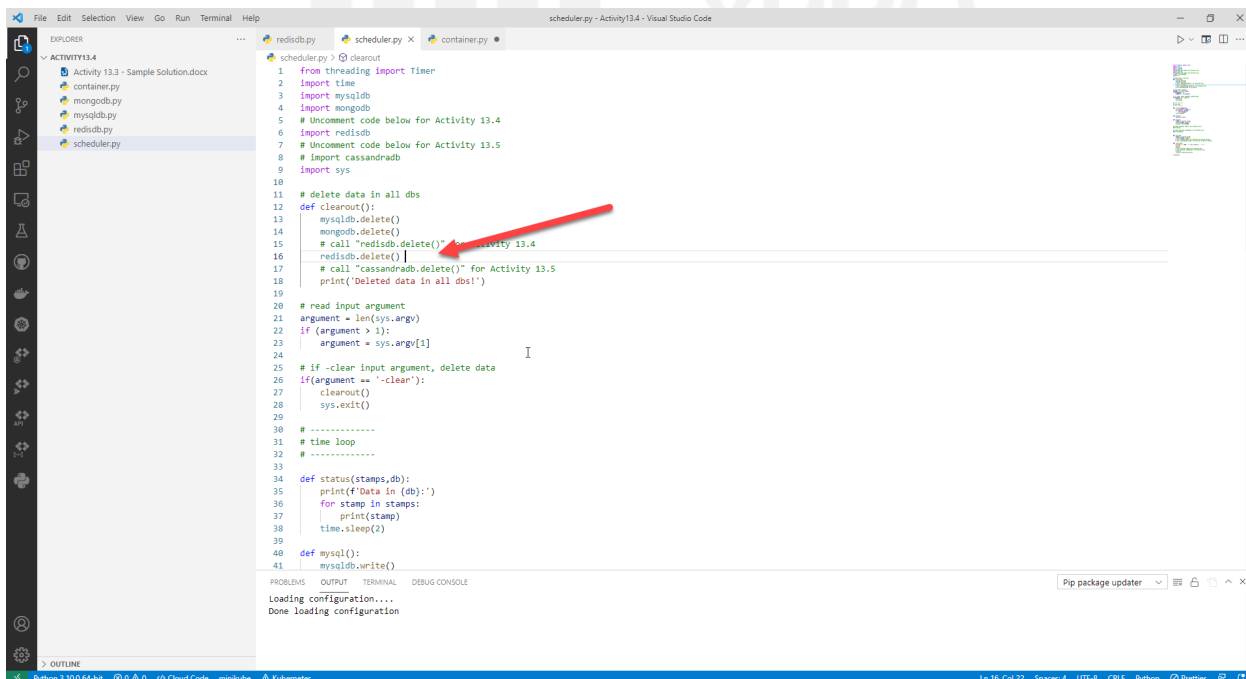


```

1  from threading import Timer
2  import time
3  import mysqldb
4  import mongod
5  # Uncomment code below for Activity 13.4
6  import redisdb
7  # Uncomment code below for Activity 13.5
8  import cassandradb
9  import sys
10
11 # delete data in all dbs
12 def clearout():
13     mysqldb.delete()
14     mongod.delete()
15     # call "redisdb.delete()" for Activity 13.4
16     # call "cassandradb.delete()" for Activity 13.5
17     print('Deleted data in all dbs!')
18
19 # read input argument
20 argument = len(sys.argv)
21 if (argument > 1):
22     argument = sys.argv[1]
23
24 # if -clear input argument, delete data
25 if (argument == '-clear'):
26     clearout()
27     sys.exit()
28
29 # -----
30 # time loop
31 # -----
32
33 def status(stamps,db):
34     print(f'Data in {db}:')
35     for stamp in stamps:
36         print(stamp)
37         time.sleep(2)
38
39 def mysql():
40     mysqldb.write()
41

```

b.

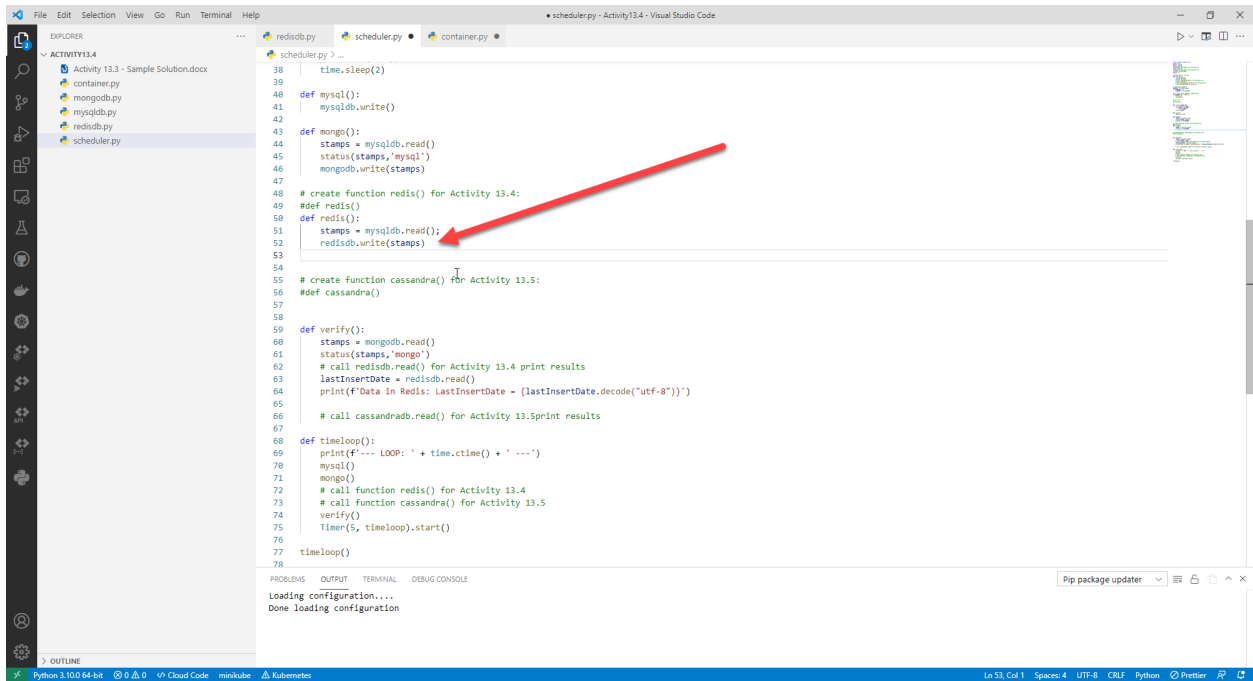


```

1  from threading import Timer
2  import time
3  import mysqldb
4  import mongod
5  # Uncomment code below for Activity 13.4
6  import redisdb
7  # Uncomment code below for Activity 13.5
8  import cassandradb
9  import sys
10
11 # delete data in all dbs
12 def clearout():
13     mysqldb.delete()
14     mongod.delete()
15     # call "redisdb.delete()" for Activity 13.4
16     redisdb.delete()
17     # call "cassandradb.delete()" for Activity 13.5
18     print('Deleted data in all dbs!')
19
20 # read input argument
21 argument = len(sys.argv)
22 if (argument > 1):
23     argument = sys.argv[1]
24
25 # if -clear input argument, delete data
26 if (argument == '-clear'):
27     clearout()
28     sys.exit()
29
30 # -----
31 # time loop
32 # -----
33
34 def status(stamps,db):
35     print(f'Data in {db}:')
36     for stamp in stamps:
37         print(stamp)
38         time.sleep(2)
39
40 def mysql():
41     mysqldb.write()
42

```

8c.

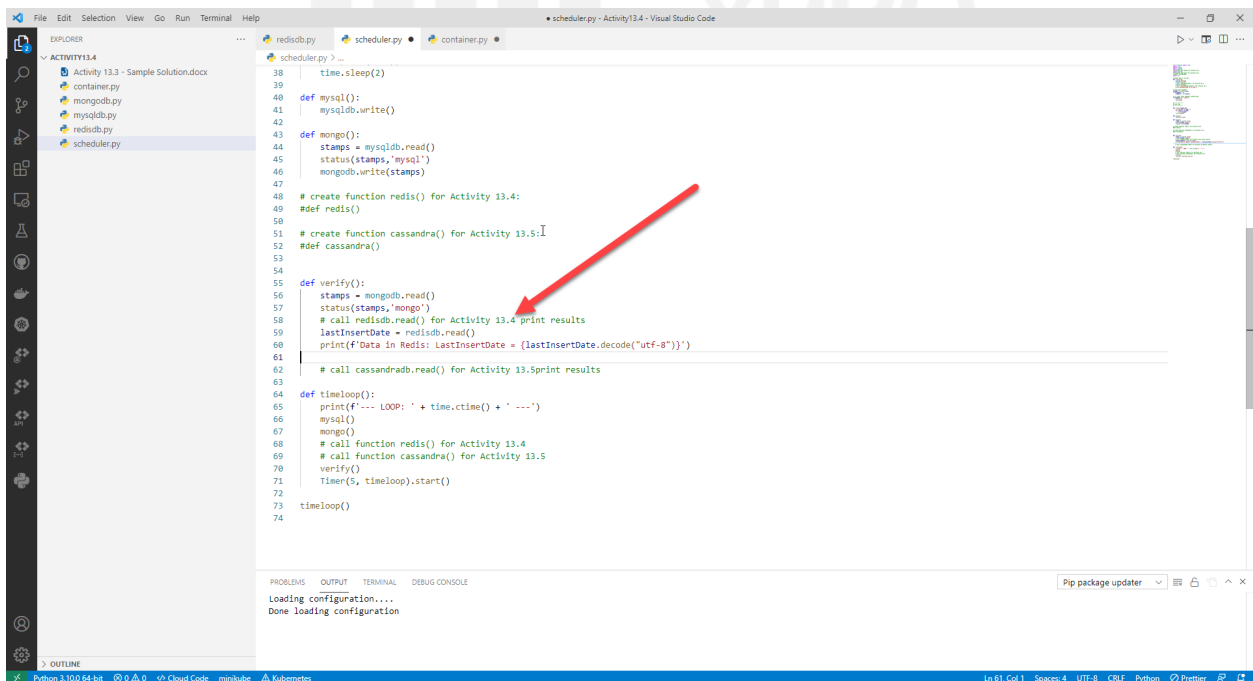


```

38 time.sleep(2)
39
40 def mysql():
41     mysql.write()
42
43 def mongo():
44     stamps = mysql.read()
45     status(stamps, 'mysql')
46     mongo.write(stamps)
47
48 # create function redis() for Activity 13.4:
49 def redis():
50     stamps = mysql.read()
51     stamps = redisdb.read()
52     redisdb.write(stamps)
53
54
55 # create function cassandra() for Activity 13.5:
56 def cassandra():
57
58
59 def verify():
60     stamps = mongo.read()
61     status(stamps, 'mongo')
62     # call redisdb.read() for Activity 13.4 print results
63     lastInsertDate = redisdb.read()
64     print(f'Data in Redis: LastInsertDate = {lastInsertDate.decode("utf-8")}')
65
66     # call cassandradb.read() for Activity 13.5 print results
67
68 def timeloop():
69     print(f'--- LOOP: ' + time.ctime() + ' ---')
70     mysql()
71     mongo()
72     # call function redis() for Activity 13.4
73     # call function cassandra() for Activity 13.5
74     verify()
75     Timer(5, timeloop).start()
76
77 timeloop()
78

```

8d.

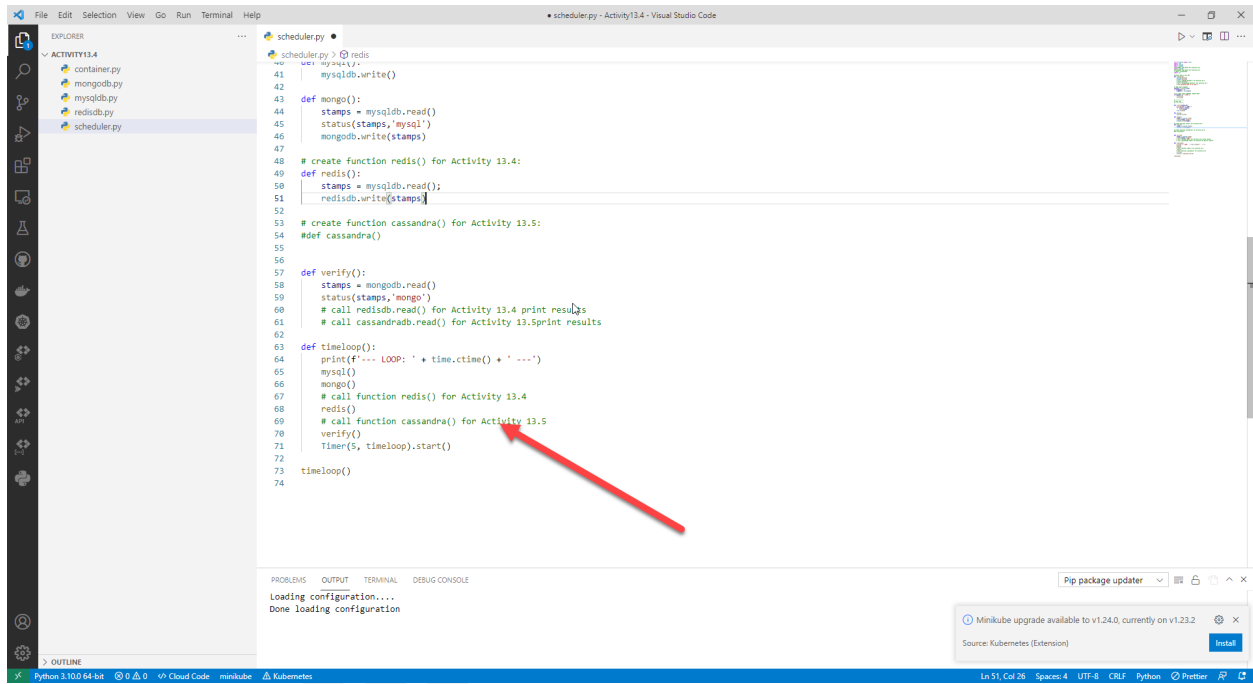


```

38 time.sleep(2)
39
40 def mysql():
41     mysql.write()
42
43 def mongo():
44     stamps = mysql.read()
45     status(stamps, 'mysql')
46     mongo.write(stamps)
47
48 # create function redis() for Activity 13.4:
49 def redis():
50
51 # create function cassandra() for Activity 13.5:
52 def cassandra()
53
54
55 def verify():
56     stamps = mongo.read()
57     status(stamps, 'mongo')
58     # call redisdb.read() for Activity 13.4 print results
59     lastInsertDate = redisdb.read()
60     print(f'Data in Redis: LastInsertDate = {lastInsertDate.decode("utf-8")}')
61
62     # call cassandradb.read() for Activity 13.5 print results
63
64 def timeloop():
65     print(f'--- LOOP: ' + time.ctime() + ' ---')
66     mysql()
67     mongo()
68     # call function redis() for Activity 13.4
69     # call function cassandra() for Activity 13.5
70     verify()
71     Timer(5, timeloop).start()
72
73 timeloop()
74

```

8e.

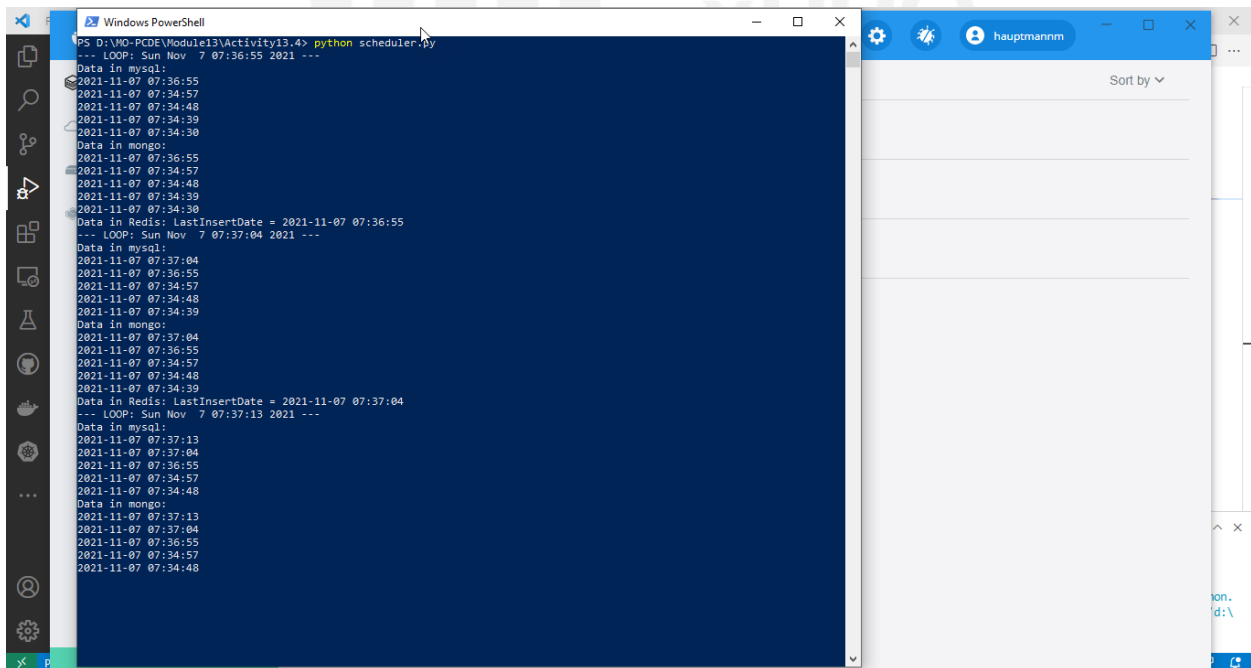


```

scheduler.py
41 mysql.write()
42
43 def mongo():
44     stamps = mysql.read()
45     status(stamps, 'mysql')
46     mongo.write(stamps)
47
48 # create function redis() for Activity 13.4:
49 def redis():
50     stamps = mysql.read()
51     redis.write(stamps)
52
53 # create function cassandra() for Activity 13.5:
54 def cassandra()
55
56
57 def verify():
58     stamps = mongo.read()
59     status(stamps, 'mongo')
60     # call redis.read() for Activity 13.4 print results
61     # call cassandra.read() for Activity 13.5 print results
62
63 def timeLoop():
64     print('--- LOOP: ' + time.time() + ' ---')
65     mysql()
66     mongo()
67     # call function redis() for Activity 13.4
68     redis()
69     # call function cassandra() for Activity 13.5
70     verify()
71     timer(5, timeLoop).start()
72
73 timeLoop()
74

```

9.

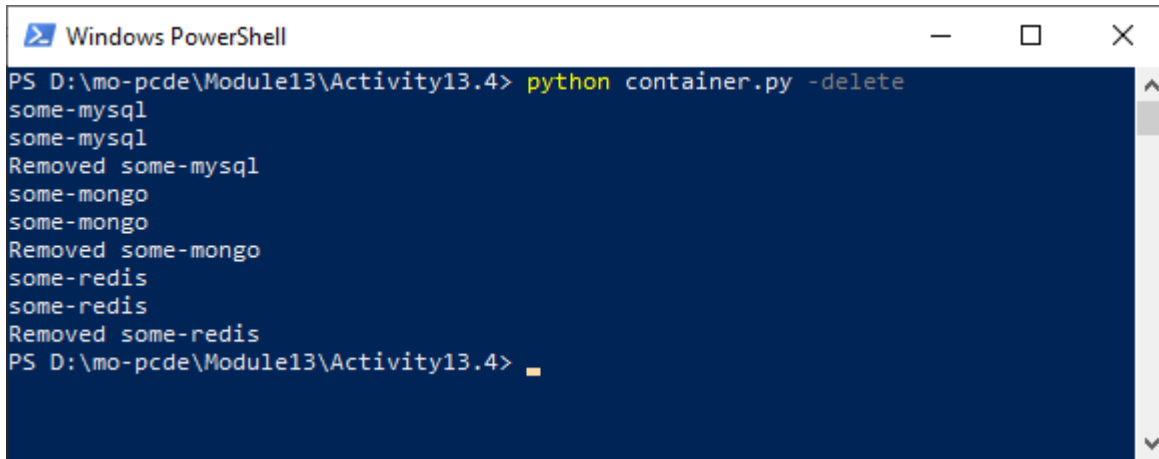


```

PS D:\VO-PCDE\Module13\Activity13.4> python scheduler.py
--- LOOP: Sun Nov 7 07:36:55 2021 ---
Data in mysql:
2021-11-07 07:36:55
2021-11-07 07:34:57
2021-11-07 07:34:48
2021-11-07 07:34:39
2021-11-07 07:34:30
Data in mongo:
2021-11-07 07:36:55
2021-11-07 07:34:57
2021-11-07 07:34:48
2021-11-07 07:34:39
2021-11-07 07:34:30
Data in Redis: LastInsertDate = 2021-11-07 07:36:55
--- LOOP: Sun Nov 7 07:37:04 2021 ---
Data in mysql:
2021-11-07 07:37:04
2021-11-07 07:36:55
2021-11-07 07:34:57
2021-11-07 07:34:48
2021-11-07 07:34:39
Data in mongo:
2021-11-07 07:37:04
2021-11-07 07:36:55
2021-11-07 07:34:57
2021-11-07 07:34:48
Data in Redis: LastInsertDate = 2021-11-07 07:37:04
--- LOOP: Sun Nov 7 07:37:13 2021 ---
Data in mysql:
2021-11-07 07:37:13
2021-11-07 07:37:04
2021-11-07 07:36:55
2021-11-07 07:34:57
2021-11-07 07:34:48
Data in mongo:
2021-11-07 07:37:13
2021-11-07 07:37:04
2021-11-07 07:36:55
2021-11-07 07:34:57
2021-11-07 07:34:48

```


10.



```
Windows PowerShell
PS D:\mo-pcde\Module13\Activity13.4> python container.py -delete
some-mysql
some-mysql
Removed some-mysql
some-mongo
some-mongo
Removed some-mongo
some-redis
some-redis
Removed some-redis
PS D:\mo-pcde\Module13\Activity13.4>
```

11.

